

Lesson 2: Fields

A map of force, not a visible substance

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

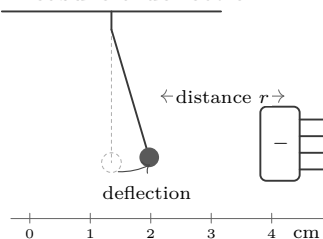
What you need

- Tape charge source
- Thread-suspended pith or foil ball
- Ruler
- Graph paper

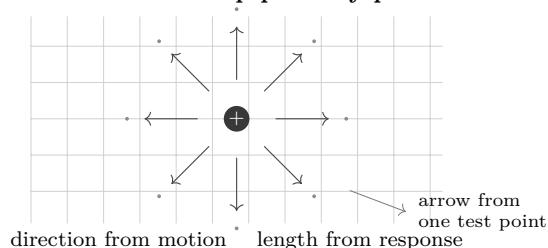
Predict first

As separation doubles, should the interaction strengthen, weaken, or remain unchanged?

A. Measure a deflection



B. Build the map point by point



Investigate

1. Suspend the lightweight test object and mark its rest position.
2. Place the prepared charge source at measured distances without touching.
3. Record deflection direction and magnitude for at least five separations.
4. Sketch arrows at each test point: direction from motion, length from relative response.

Explain from the evidence

A field is a model assigning an interaction to each location. An electric-field arrow points in the direction a positive test charge would accelerate. Field lines are a drawing convention: they do not begin to exist because ink connects the arrows, and crossings would assign two directions to one point.

Change one thing

Draw the map first from prediction, then overlay the observations in a different line style. Circle disagreements and propose a test.

Evidence record

Prediction

One change

Observation

Claim + because

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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